3 Identities and inequalities

What students need to learn	Notes
A Simple algebraic division	Division by $(x + a)$, $(x - a)$, $(ax + b)$ or $(ax - b)$ will be required.
B The factor and remainder theorems	Students should know that if $f(x) = 0$ when $x = a$, then $(x - a)$ is a factor of $f(x)$. Students may be required to factorise cubic expressions such as: $x^3 + 3x^2 - 4$ and $6x^3 + 11x^2 - x - 6$, when a factor has been provided. Students should be familiar with the terms 'quotient' and 'remainder' and be able to determine the remainder when the polynomial $f(x)$ is divided by $(ax + b)$ or $(ax - b)$.
C Solutions of equations, extended to include the simultaneous solution of one linear and one quadratic equation in two variables	The solution of a cubic equation containing at least one rational root may be set.
D Simple inequalities, linear and quadratic	For example $ax + b > cx + d$, $px^2 + qx + r < sx^2 + tx + u$
E The graphical representation of linear inequalities in two variables	The emphasis will be on simple questions designed to test fundamental principles. Simple problems on linear programming may be set.